Sahaj Bhadja

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Summary

Software Engineer with 2+ years building reliable applications using **C#/.NET**, **C++**, object-oriented design, **asynchronous programming**, user interfaces, automated testing, and cross-functional delivery for control-oriented systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Software Engineering, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Computer Networks, Database Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Restructured a fault-tolerant **asynchronous programming** workflow with FastAPI workers, explicit job states, and failure isolation, tripling throughput while processing 1K+ concurrent AI jobs without cascading failures in production.
- Designed normalized Azure SQL schemas and indexes for 500+ users, cutting dashboard latency from 1.2 seconds to 350 milliseconds while sustaining 99%+ **application reliability** across 5K+ monthly jobs.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Developed concurrent file-processing services with multithreading, secure file ingestion, and fault-tolerant execution, reliably completing 1K+ files per batch while applying **Producer/Consumer** workflows to control peak demand safely.
- Delivered React and Flask interfaces for uploads, grading history, and feedback, supporting 150+ instructors and 10K+ daily requests while validating releases through **integration testing** and CI/CD deployments.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a Decision Transformer for autonomous-driving reinforcement learning, improving trajectory planning 18% while processing 100K+ simulated driving logs through reproducible **software validation** pipelines for control-oriented research experiments.
- Architected a spatial-temporal encoder combining HD-map geometry and ego trajectories through multi-head attention, improving closed-loop stability 15% across **simulation testing** baselines and documented rigorous autonomous-driving research evaluations.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Designed cross-platform modules using **object-oriented design patterns**, including Singleton and Factory, improving API response times 30%, reducing technical debt 20%, and increasing maintainability across a microservice data-sharing platform.
- Automated 20+ **unit and integration tests** with PyTest, Mockito, Jenkins, and virtual-machine builds across Windows, Linux, and macOS while collaborating with Microsoft and GitHub stakeholders during Agile delivery.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight **C# and ASP.NET Core** microservices with Entity Framework and Azure SQL for authentication, resources, billing, permissions, and notifications, synchronizing 2M+ records across distributed locations in real time.
- Developed a React and TypeScript **user interface** with real-time WebSocket updates and REST integrations; Jenkins CI/CD, Docker staging, and automated tests cut deployment time 50% across releases.

Skills

Languages: C#, C++, Python, Java, TypeScript, JavaScript, SQL, HTML5
Frameworks: .NET, ASP.NET Core, WPF, MVVM, React, FastAPI, Flask, Entity Framework, REST APIs
Design & Systems: OOP, SOLID, Design Patterns, Async/Await, Multithreading, TCP/IP, UDP, Linux, Windows
Data & Cloud: Azure SQL, SQL Server, PostgreSQL, MySQL, Microsoft Azure, AWS, Blob Storage
Quality: Unit Testing, Integration Testing, Debugging, Git, Jenkins, CI/CD, Docker, Agile/Scrum, Documentation, Technical Communication

Projects

Instrument Control Workbench | *C#, .NET 8, WPF, MVVM, Async/Await, TCP/IP, UDP, xUnit*

- Designed a desktop **hardware control** application with asynchronous command queues, TCP/IP and UDP device adapters, live telemetry, cancellation handling, and unit-tested MVVM components for repeatable laboratory workflows.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2*

- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel-queue **scheduling algorithms** in C++ with React controls, validating 1K+ process combinations and reducing average turnaround time up to 25%.

Chat2SQL Code | *Java, TypeScript, SQL, Docker, Maven, Node.js*

- Extended a cross-platform Java and TypeScript workspace supporting 10+ databases, using **object-oriented architecture**, schema synchronization, validated query execution, and Docker environments to streamline development across operating systems.